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Data Creates Competitive Advantage – Probably Not Always.

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Access to consumer data has brought about a focal change in the business environment, from small scale industries to tech giants, the synthesis of consumer information to make it relevant to the business

operations, and the curation of products became a priority. Interestingly, the storage of data dates back to the 18,000 BCE, when the Palaeolithic tribespeople would mark notches into sticks or bones to keep a track of trading activities. Now, most firms use consumer data to create an unbeatable competitive advantage in the market – the more customers a firm has, the greater the data gathering process is. However, more often than not, firms grossly overestimate their ability to gain a competitive advantage through consumer data which we shall be talking about in detail.

Competitive advantage refers to factors that allow a company to produce goods or services better or more cheaply than its rivals, allowing the firm to generate more sales or superior margins compared to its market rivals. One of the most important concepts to understand competition in a market is “Value Chain”. A value chain involves two categories of activities termed as the “value activities”, that may be technological or economical. To remain profitable business must create more value than its cost of operations, and to gain a competitive advantage a business must either perform these activities at a lower cost or perform them in a way that leads to differentiation and a higher price. Data might provide the firms with a competitive advantage, but it does not act as a complete barrier to entry or replication.

The value gained from collecting data and using data-enabled learning refers to the amount of benefit received from it. The firms need to decipher the point beyond where the customer data becomes irrelevant and no longer provides an additional benefit. When the marginal value of learning from customer data remains high, even after a very large customer base has been acquired, products and services tend to have significant competitive advantages. The value of data specific activities can usually be ascertained by the amount the customer’s willingness to pay for the product. Some firms can charge a higher price for their product or services based on the value proposition created by them in the market based on the analysis of customers, for example, the song

recommendation and daily playlist services on Spotify, or personalized recommendations on Netflix.

One of the major issues with firms that are dependent on consumer data is that this data can be purchased or reverse engineered, thus reducing their competitive advantage because of the possibility of replication of their product. It's important to keep in mind that technological progress can undermine a position based on unique or proprietary data.

As a result, it becomes very essential to have access to unique customer data with fewer or no substitutes. An example can be found in the market for speech recognition software. For a long period, this market was dominated by Nuance's Dragon solutions, as the user had to train the software to understand their tones and speech patterns. With the independent speech-recognition systems, this software takes little to no time to imitate the same behaviour making this technology extremely accessible in the market and destroying Nuance's leverage in the market. Here arises the question regarding the durability of the competitive advantage, as the concern as to how sustainable it is. For this, companies need to consider two factors – firstly, how frequently the insights from customer data change and secondly, whether the improvements are hidden or deeply embedded in a complex production process, making them hard to replicate.

In a market, purchasing data is more accessible than acquiring customers – making it simpler for everyone to access the same pool of resources. To assemble lasting data-enabled network effects, the firm has to work invariably to learn from customer data. In many cases nearly all the profits of learning from customer data can be achieved with relatively low numbers of customers, and dramatic advancements in AI will reduce the need for client data to the point where the value of data-enabled learning might dissolve completely. The future of the market would involve regular network effects and enhanced by data-

enabled learning, to maintain a sustainable competitive advantage, in the light of recent advancements.

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